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A Directed-Hypergraph Representation of Higher-Order Information Flows

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Abstract

A statistical approach to higher-order interactions requires an understanding of higher-order correlations among components. The information dynamics of these correlations can be encoded using multiple transfer entropy and multiple reversed transfer entropy. This paper investigates the relationship between the topology of higher-order information flows and the entropy dynamics of individual components. Because a multiple transfer entropy has a source set and a target set, its structure can be naturally represented by a hyperarc. By distinguishing statistically direct and indirect relationships, we introduce an information flow hypergraph and analyze its BF-hypergraph substructure in terms of BF-sinks, BF-sources, and weak BF-hypercycles. Under the strong blocking flows condition and $\alpha_1 = 0$, nontrivial oscillatory or recurrent information dynamics requires the presence of a weak BF-hypercycle. We further show that stationarity imposes an additional local balance condition between B-inflows and F-outflows. The corresponding result for cycles in classical directed graph theory is recovered as a special case.

1 Introduction

Synergistic effects are key characteristics of higher-order dynamics [1, 2, 3, 4] and distinguish many-body interactions from purely pairwise interactions. For a statistical characterization of such dynamics, several studies [5, 6, 7, 8] have adopted information-theoretic approaches. In particular, partial information decomposition (PID) [8, 9] provides nonnegative measures of synergistic and redundant information, although their definitions are not unique. In this paper, we instead employ multiple mutual information, also referred to as interaction information or co-information [10, 11, 12, 13, 14, 5, 15, 16]. Although it is a signed quantity, multiple mutual information is uniquely defined consistently for arbitrary orders. It therefore captures higher-order correlation structures that cannot be reduced to ordinary pairwise mutual information [5, 6].

The temporal evolution of such higher-order correlations can be described using multiple transfer entropy and multiple reversed transfer entropy [17]. These quantities extend the decomposition of mutual-information dynamics to higher orders, capturing directional higher-order information flows. However, their topological organization has not yet been systematically characterized.

This paper introduces an information flow hypergraph to represent the topology of direct higher-order information flows. A multiple transfer entropy $T^{n+m-1}_{\{a_1,\ldots,a_n\}\to\{b_1,\ldots,b_m\}}$ contains a source set and a target set, which naturally correspond to the tail and head of a hyperarc [18, 19, 20]. Since nonzero transfer entropy may also reflect indirect statistical dependencies, direct information flows are identified using dynamical neighbors. The resulting representation consists of hyperarcs corresponding to direct multiple transfer entropies and multiple reversed transfer entropies.

The entropy dynamics of an individual vertex is governed by a BF-hypergraph substructure composed of B-arcs and F-arcs [18, 19, 21]. Under the strong blocking flows condition and deterministic reversible dynamics, this structure constrains the possible evolution of the vertex entropy. In particular, a finite oriented BF-hypergraph without weak BF-hypercycles possesses a BF-sink and a BF-source. At a BF-sink, the entropy is nondecreasing and increases strictly when a nonzero B-inflow is present. This result generalizes the pairwise argument developed in Ref. [22].

A weak BF-hypercycle is therefore necessary for oscillatory or recurrent information dynamics under the standing assumptions. However, unlike a cycle in classical directed graph theory [23, 24], a weak BF-hypercycle does not necessarily admit a stationary state. This paper derives an additional necessary condition for stationarity: a B-inflow must be accompanied by an F-outflow, whereas the absence of a B-inflow precludes an F-outflow.

The remainder of this paper is organized as follows. Section 2 reviews the definitions of higher-order information dynamics and directed hypergraphs. Section 3 introduces the information flow hypergraph and the blocking flows conditions. Section 4 analyzes BF-hypergraphs without weak BF-hypercycles. Section 5 derives a necessary condition for stationary information dynamics on a weak BF-hypercycle.

2 Definitions

2.1 Higher-Order Information Dynamics

Consider an N-component system with the state $(\Gamma_1(t), \ldots, \Gamma_N(t))$ at time t. An ensemble description or stochastic dynamics induces a joint probability $P(\Gamma_1(t), \ldots, \Gamma_N(t))$. Information-theoretic quantities can then be used to characterize statistical correlations and directional information flows among the components.

Definition 1 *The* ***Shannon entropy*** H_i *and* ***conditional entropy*** $H_j|_i$ *[25, 26] are defined as:*

$$\begin{aligned} H_i &:= -\sum_{\gamma_i \in \mathcal{M}} P(\Gamma_i = \gamma_i) ln\ P(\Gamma_i = \gamma_i) \\ H_j|_i &:= H_{ij} - H_i, \end{aligned} \tag{1}$$

where Γ_i *and* Γ_j *are discrete random variables in a state space* $\mathcal{M}$*.*

Definition 2 *The* ***mutual information*** *[26] between* Γ_i *and* Γ_j *quantifies the reduction in the uncertainty of* Γ_j *obtained by conditioning on* Γ_i*:*

$$I_{i:j} := H_j - H_j|_i = H_i + H_j - H_{ij}. \tag{2}$$

The ***conditional mutual information*** *is defined as*

$$I_{i:j}|_k := H_i|_k + H_j|_k - H_{ij}|_k. \tag{3}$$

For discrete random variables, the Shannon entropy, conditional entropy, and mutual information are nonnegative.

Definition 3 *The* ***multiple mutual information*** *[10, 11, 13] is a signed generalization of mutual information that characterizes higher-order statistical dependence.*

$$I^n_{i_1:i_2:\ldots:i_{n+1}} := \sum_{r=1}^{n+1} (-1)^{r+1} \sum_{(j_1,\ldots,j_r)\in Y^{n+1}_r} H_{j_1,\ldots,j_r}, \tag{4}$$

where the index set $Y^{n+1}_l := \{(j_1, j_2, \ldots, j_l)|j_1, j_2, \ldots, j_l \ \in \{i_1, i_2, \ldots, i_{n+1}\} \text{ and } j_1 < j_2 < \cdots < j_l\}$ *for* $n \in \mathbb{N}$*.*

In this notation, the superscript n denotes the interaction order, and I^n involves $n+1$ random variables.

The temporal evolution of correlations can be related to directional information flows [17]. In an information-theoretic framework, such directional dependencies are characterized using transfer entropy [27].

Definition 4 *In the one-step formulation adopted here, the* ***transfer entropy*** *[28, 29] from* j *to* i *measures a causal impact of* j *on the dynamics of* i*:*

$$T_{j\to i} := I_{i':j}|_i, \tag{5}$$

where unprimed and primed variables are evaluated at times t *and* $t + \Delta t$*, respectively, with* $\Delta t > 0$*.*

Definition 5 ***Reversed transfer entropy*** *[30, 31, 32], also referred to as backward transfer entropy [30], is defined by reversing the temporal indexing in the transfer-entropy expression:*

$$rT_{j\to i} := I_{i:j'}|_{i'}. \tag{6}$$

Definition 6 ***Multiple transfer entropy*** *and* ***multiple reversed transfer entropy*** *[17] are higher-order generalizations of transfer entropy and reversed transfer entropy:*

$$\begin{aligned} T^n_{(i)_m \to (j)_l} &:= I^n_{j'_1:\cdots:j'_l:i_1:\cdots:i_m}|_{j_1,\ldots,j_l} \\ rT^n_{(i)_m \to (j)_l} &:= I^n_{j_1:\cdots:j_l:i'_1:\cdots:i'_m}|_{j'_1,\ldots,j'_l}, \end{aligned} \tag{7}$$

where $n = m + l - 1$, $(i)_m := \{i_1, i_2, \ldots, i_m\}$, *and* $(j)_l := \{j_1, j_2, \ldots, j_l\}$.

Using these quantities, Ref. [17] derived exact decompositions of the temporal changes in entropy, mutual information, and multiple mutual information:

$$\begin{aligned} \Delta H_i &= \sum_{\eta \in K_i} [T_{\eta\to i} - rT_{\eta \to i}] - E_{K_i \to i} + \alpha_1^{K_i \to i} \\ \Delta I_{i:j} &= [T_{j\to i} - rT_{j\to i}] + [T_{i\to j} - rT_{i \to j}] + \alpha_2^{ij} \\ \Delta I^n_{\eta_1:\cdots:\eta_{n+1}} &= \sum_{((i)_m,(j)_l)\in D_{S_\eta}} [T^n_{(i)_m\to(j)_l} - rT^n_{(i)_m\to(j)_l}] + \alpha_2^{S_\eta}, \end{aligned} \tag{8}$$

where $D_{S_\eta} := \{(U, V)|U \neq \emptyset, V \neq \emptyset, U \cup V = S_\eta, U \cap V = \emptyset\}$ and,

$$\begin{aligned} E_{K_i\to i} &:= \sum_{u=2}^{m} (-1)^u \sum_{(\eta)_u \subset K_i} [T^u_{(\eta)_u \to i} - rT^u_{(\eta)_u\to i}] \\ \alpha_1^{K_i\to i} &:= H_{i'}|_{i\ \zeta_1\ \ldots\ \zeta_m} - H_i|_{i'\ \zeta'_1\ \ldots\ \zeta'_m} \\ \alpha_2^{ij} &:= I_{i':j'}|_{i\ j} - I_{i:j}|_{i'\ j'} \\ \alpha_2^{S_\eta} &:= I^n_{\eta'_1:\cdots:\eta'_{n+1}}|_{\eta_1\ \ldots\ \eta_{n+1}} - I^n_{\eta_1:\cdots:\eta_{n+1}}|_{\eta'_1\ \ldots\ \eta'_{n+1}}, \end{aligned} \tag{9}$$

where $K_i := \{\zeta_1, \zeta_2, ..., \zeta_{m_i}\}$ and $S_\eta = \{\eta_1, \eta_2, ..., \eta_{n+1}\}$.

The terms α_1 and α_2 represent residual contributions that are not expressed as transfer entropy or reversed transfer entropy. Their roles under deterministic reversible dynamics will be discussed in Section 3.3.

2.2 Directed Hypergraphs

We summarize the definitions from directed-hypergraph theory that are required for the subsequent analysis [18, 19, 20, 33].

Definition 7 *A* ***directed hypergraph*** $(V, \overrightarrow{E})$ *consists of a finite set* V *of vertices and a nonempty set* $\overrightarrow{E} \subset \{(e^+, e^-)|e^+, e^- \subset V$ *and* $e^+ \neq \emptyset, e^- \neq \emptyset, e^+ \cap e^- = \emptyset\}$ *of hyperarcs. The set of reversed hyperarcs is defined as* $\overleftarrow{E} := \{(e^-, e^+)|(e^+, e^-) \in \overrightarrow{E}\}$. *The directed hypergraph* $(V, \overleftarrow{E})$ *is called the* ***symmetric image*** *of* $(V, \overrightarrow{E})$. *For a hyperarc* $\overrightarrow{e} = (e^+, e^-) \in \overrightarrow{E}$, *the sets* e^+ *and* e^- *are called its tail and head, respectively, and are denoted by* $\mathcal{T}(\overrightarrow{e})$ *and* $\mathcal{H}(\overrightarrow{e})$.

We assume that a directed hypergraph $(V, \overrightarrow{E})$ has no isolated vertices, namely,

$$V = \bigcup_{\overrightarrow{e}\in\overrightarrow{E}} [\mathcal{T}(\overrightarrow{e}) \cup \mathcal{H}(\overrightarrow{e})]. \tag{10}$$

Definition 8 *An* ***oriented hypergraph*** $(V, \overrightarrow{E})$ *is a directed hypergraph* $(V, \overrightarrow{E})$ *satisfying* $\overrightarrow{E} \cap \overleftarrow{E} = \emptyset$.

Definition 9 *A* ***BF-hypergraph*** $(V, \overrightarrow{E})$ *is a directed hypergraph* $(V, \overrightarrow{E})$ *such that* $|e^+| = 1$ *or* $|e^-| = 1$ *for every* $(e^+, e^-) \in \overrightarrow{E}$, *and* $V = \bigcup_{\overrightarrow{e}\in\overrightarrow{E}}[\mathcal{T}(\overrightarrow{e}) \cup \mathcal{H}(\overrightarrow{e})]$. *Specifically, the hyperarc* $\overrightarrow{e_B} = (e^+, e^-) \in \overrightarrow{E}$ *with* $|e^-| = 1$ *is called the B-arc, while the hyperarc* $\overrightarrow{e_F} = (e^+, e^-) \in \overrightarrow{E}$ *with* $|e^+| = 1$ *is called the F-arc. An ordinary directed edge in a directed graph satisfies both conditions and can therefore be regarded as both a B-arc and an F-arc.*

Definition 10 *For a vertex* $v \in V$ *in a BF-hypergraph* $(V, \overrightarrow{E})$, *the* ***B-inflow*** *set of* v, *denoted by* $In_B(v)$, *is the set of B-arcs directed toward* v: $In_B(v) := \{(e^+, e^-) \in \overrightarrow{E}|e^- = \{v\}\}$. *Similarly, the* ***B-outflow*** *set of* v *is defined as* $Out_B(v) := \{(e^+, e^-) \in \overrightarrow{E}|v \in e^+$ *and* $|e^-| = 1\}$. *The definitions for* ***F-inflow*** *and* ***F-outflow*** *are* $In_F(v) := \{(e^+, e^-) \in \overrightarrow{E}|v \in e^-$ *and* $|e^+| = 1\}$ *and* $Out_F(v) := \{(e^+, e^-) \in \overrightarrow{E}|e^+ = \{v\}\}$.

Definition 11 *A **BF-sink** is a vertex $v \in V$ such that $Out_B(v) = \emptyset$, $Out_F(v) = \emptyset$, and $In_B(v) \cup In_F(v) \neq \emptyset$. A **BF-source** is a vertex $v \in V$ such that $In_B(v) = \emptyset$, $In_F(v) = \emptyset$, and $Out_B(v) \cup Out_F(v) \neq \emptyset$.*

Definition 12 *In a directed hypergraph $(V, \overrightarrow{E})$, a **weak hyperpath** from v_1 to v_{n+1} is defined as a $(2n+1)$-tuple $(v_1, \overrightarrow{e_1}, v_2, \overrightarrow{e_2}, \ldots, \overrightarrow{e_n}, v_{n+1})$ such that $v_i \in \mathcal{T}(\overrightarrow{e_i})$ and $v_{i+1} \in \mathcal{H}(\overrightarrow{e_i})$ with $\overrightarrow{e_i} \in \overrightarrow{E}$ for $\forall i \in \{1, 2, \ldots, n\}$. A vertex $s \in V$ is **connected** to $u \in V$ if a weak hyperpath from u to s exists. Furthermore, a nonempty subset $S \subset V$ is **connected** to the nonempty subset $U \subset V$ if there are $u \in U$ and $s \in S$ such that s is connected to u.*

Definition 13 *The **transitive closure** of a directed hypergraph $\mathcal{G} = (V, \overrightarrow{E})$ is a directed hypergraph $\mathcal{G}^* = (V, \overrightarrow{E^*})$ such that, for any nonempty disjoint subsets $U, S \subset V$, $(U, S) \in \overrightarrow{E^*}$ if and only if S is connected to U in $\mathcal{G}$.*

Definition 14 *A **weak BF-hypercycle** of length n in a BF-hypergraph $(V, \overrightarrow{E}_{BF})$ is a weak hyperpath $(v_1, \overrightarrow{e_1}, v_2, \overrightarrow{e_2}, \ldots, \overrightarrow{e_n}, v_{n+1})$ in $(V, \overrightarrow{E}_{BF})$ such that $v_1 = v_{n+1}$.*

In the definition of weak BF-hypercycle, no distinctness condition is imposed on the intermediate vertices or hyperarcs.

3 Identification of Higher-Order Information Flows

3.1 Information Flow Hypergraph

Consider an N-component system with a joint probability distribution. Let $V = \{v_1, \ldots, v_N\}$ be the set of components. The set of all nonzero multiple transfer entropies is defined as $\overrightarrow{E}_{total} := \{((j)_m, (i)_l) | (j)_m, (i)_l \subset V$ and $T^{m+l-1}_{(j)_m \to (i)_l} \neq 0\}$. However, $\overrightarrow{E}_{total}$ may contain indirect statistical relationships. For example, $T_{p \to q}$ can be nonzero even when the influence of p reaches q through intermediate components. Therefore, we effectively consider direct causal relationships using dynamical neighbors.

Definition 15 *A set of **dynamical neighbors** $K_{v_1,\ldots,v_l} := \{w_1, \ldots, w_m\} \subset V$ of the components $v_1, \ldots, v_l$ is an inclusion-minimal set satisfying $H_{v'_1,\ldots,v'_l} |_{v_1,\ldots,v_l,w_1,\ldots,w_m} = 0$, where the primed variables are evaluated at the next time step.*

To ensure that the dynamical neighbors are well defined, we assume the dynamics of N components is described as:

$$(\Gamma_{v_1}(t + \Delta t), \ldots, \Gamma_{v_N}(t + \Delta t)) = F(\Gamma_{v_1}(t), \ldots, \Gamma_{v_N}(t), t), \tag{11}$$

where Γ_i is the state of component i. Stochastic dynamics can be represented in the form of Eq. (11) by introducing additional noise-source vertices.

However, the set of dynamical neighbors is not necessarily unique. We therefore define $\overline{\mathcal{K}}_{v_1,\ldots,v_l} := \{K \subset V | K$ is a set of dynamical neighbors of $v_1, \ldots, v_l\}$.

Then, the set of direct multiple transfer entropies is defined as $\overrightarrow{E}_T := \{((j)_m, (i)_l) \in \overrightarrow{E}_{total} | \exists K_{(i)_l} \in \overline{\mathcal{K}}_{(i)_l}$ such that $(j)_m \subset K_{(i)_l}\}$. The set $\overrightarrow{E}_T$ is more efficient to calculate α_1 in Eq. (8) than $\overrightarrow{E}_{total}$, as discussed in Sec. 3.3. However, Proposition 1 shows that the connectivity of $(V, \overrightarrow{E}_{total})$ is not generally preserved in $(V, \overrightarrow{E}_T)$. Thus, some indirect statistical relationships cannot be represented as chains of direct information flows.

Proposition 1 *In general, $\mathcal{G}^*_T \neq \mathcal{G}^*_{total}$, where $\mathcal{G}_T = (V, \overrightarrow{E}_T)$, $\mathcal{G}_{total} = (V, \overrightarrow{E}_{total})$.*

(proof)
Appendix A.

However, if one modifies the definition of $\overrightarrow{E}_T$ as $\overrightarrow{E}_{\bar{T}} := \{((j)_m, (i)_l) \in \overrightarrow{E}_{total} | \exists K_{(i)_l} \in \overline{\mathcal{K}}_{(i)_l}$ such that $(j)_m \cap K_{(i)_l} \neq \emptyset\}$, the specific example for Proposition 1, addressed in Appendix A, loses its meaning: X and Y are connected in $\mathcal{G}_{\bar{T}} = (V, \overrightarrow{E}_{\bar{T}})$, as the redundant information flow exists. Specifically, $T^2_{\{Y,C\} \to \{X\}} = ln\ 2$, indicating $(\{Y, C\}, \{X\}) \in \overrightarrow{E}_{\bar{T}}$. Moreover, Proposition 2 shows that $\mathcal{G}^*_{\bar{T}} = \mathcal{G}^*_{total}$.

Proposition 2 *$\mathcal{G}^*_{\bar{T}} = \mathcal{G}^*_{total}$, where $\mathcal{G}_{\bar{T}} = (V, \overrightarrow{E}_{\bar{T}})$, $\mathcal{G}_{total} = (V, \overrightarrow{E}_{total})$.*

(proof)

Appendix B.

Nevertheless, this paper does not use this neighbor-anchored hyperarc set $\overrightarrow{E}_{\bar{T}}$ in the subsequent arguments.

Definition 16 *A **transfer-entropy hypergraph** $\mathcal{G}_T$ of an N-component system is a directed hypergraph $(V, \overrightarrow{E}_T)$, where V is a set of N components $V = \{v_1, \dots, v_N\}$ and $\overrightarrow{E}_T$ is the set of direct multiple transfer entropies $\overrightarrow{E}_T := \{((j)_m, (i)_l) | (j)_m, (i)_l \subset V,\ T^{m+l-1}_{(j)_m \to (i)_l} \neq 0$ and $\exists K_{(i)_l} \in \overline{\mathcal{K}}_{(i)_l}$ such that $(j)_m \subset K_{(i)_l}\}$. Similarly, a **reversed-transfer-entropy hypergraph** $\mathcal{G}_{rT}$ is a directed hypergraph $(V, \overrightarrow{E}_{rT})$, where $\overrightarrow{E}_{rT}$ is the set of direct multiple reversed transfer entropies $\overrightarrow{E}_{rT} := \{((j)_m, (i)_l) | (j)_m, (i)_l \subset V,$ $rT^{m+l-1}_{(j)_m \to (i)_l} \neq 0$ and $\exists K_{(i)_l} \in \overline{\mathcal{K}}_{(i)_l}$ such that $(j)_m \subset K_{(i)_l}\}$.*

Definition 17 *The **information flow hypergraph** of an N-component system is the ordered pair $\mathcal{G}_{\Delta I} := (\mathcal{G}_T, \mathcal{G}_{rT})$, where $\mathcal{G}_T$ and $\mathcal{G}_{rT}$ are the transfer-entropy and reversed-transfer-entropy hypergraphs, respectively.*

3.2 Blocking Flows Conditions

To exclude a trivial cycle formed by simultaneously allowing a hyperarc and its reverse, $v_1 \to (e^+, e^-) \to v_2 \to (e^-, e^+) \to v_1$, we impose the blocking flows condition introduced in Ref. [17]. This condition assigns a fixed direction to each information-flow channel.

Definition 18 *An information flow hypergraph with **the blocking flows condition** is an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ satisfying $\overrightarrow{E}_T \cap \overleftarrow{E}_T = \emptyset$, $\overrightarrow{E}_{rT} \cap \overleftarrow{E}_{rT} = \emptyset$, and $\overrightarrow{E}_T \cap \overrightarrow{E}_{rT} = \emptyset$, where $\mathcal{G}_T = (V, \overrightarrow{E}_T)$ and $\mathcal{G}_{rT} = (V, \overrightarrow{E}_{rT})$.*

$\mathcal{G}_T$ and $\mathcal{G}_{rT}$ in $\mathcal{G}_{\Delta I}$ with the blocking flows condition are oriented hypergraphs. Near a stationary state with $\Delta I^n_{\eta_1:\dots:\eta_{n+1}} = 0$, with $\alpha_2 = 0$, multiple transfer entropy and multiple reversed transfer entropy must balance in Eq. (8). Motivated by this balance, we impose a stronger structural condition on the information flow hypergraph.

Definition 19 *An information flow hypergraph with **the strong blocking flows condition** is an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ with the blocking flows condition satisfying $\overrightarrow{E}_{rT} = \overleftarrow{E}_T$, where $\mathcal{G}_T = (V, \overrightarrow{E}_T)$ and $\mathcal{G}_{rT} = (V, \overrightarrow{E}_{rT})$.*

Then, under the strong blocking flows condition, the hypergraph $\mathcal{G}_{rT}$ becomes the symmetric image of $\mathcal{G}_T$.

3.3 Condition for $\alpha_1 = 0$

The subsequent analysis assumes the condition $\alpha_1^{K_v \to v} = 0$, where $K_v := \{w_1, \dots, w_m\}$ is a set of dynamical neighbors of v. This condition is satisfied under deterministic and component-wise reversible dynamics.

By the definition of K_v, $H_{v'}|_{v, w_1, \dots, w_m} = 0$. Furthermore, suppose that,

$$\exists G \text{ such that } \Gamma_v(t) = G(\Gamma_v(t+\Delta t), \Gamma_{w_1}(t+\Delta t), \dots, \Gamma_{w_m}(t+\Delta t), t+\Delta t). \tag{12}$$

Then, $H_v|_{v', w'_1, \dots, w'_m} = 0$, and therefore,

$$\alpha_1^{K_v \to v} := H_{v'}|_{v, w_1, \dots, w_m} - H_v|_{v', w'_1, \dots, w'_m} = 0. \tag{13}$$

3.4 Example: A Single B-arc

Consider a three-component system a, b, and p with the dynamics:

$$\begin{aligned} &\Gamma_a(t+\Delta t) \sim B(\frac{1}{2}) \\ &\Gamma_b(t+\Delta t) \sim B(\frac{1}{2}) \\ &\Gamma_p(t+\Delta t) = \Gamma_a(t) \oplus \Gamma_b(t), \end{aligned} \tag{14}$$

where the updates of Γ_a and Γ_b are independent, and $B(q)$ denotes Bernoulli distribution with probability q, and $\oplus$ denotes the XOR operation.

Then,

$$\begin{aligned} &T_{a\to p} = T_{b\to p} = 0 \\ &rT_{p\to a} = rT_{p\to b} = 0 \\ &T^2_{ab\to p} = -ln\ 2 \\ &rT^2_{p\to ab} = -ln\ 2 \\ &T^2_{p\to ab} = rT^2_{ab\to p} = 0. \end{aligned} \tag{15}$$

Therefore, from the sets $V = \{a, b, p\}$, $\mathcal{G}_T = (V, \{(\{a, b\}, \{p\})\})$, and $\mathcal{G}_{rT} = (V, \{(\{p\}, \{a, b\})\})$, the information flow hypergraph of this system is defined as $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ (Fig. 1(a)), and satisfies the strong blocking flows condition.

3.5 Example: A Single F-arc

Consider a four-component system p, ζ, d, and e with the dynamics:

$$\begin{aligned} &\Gamma_p(t+\Delta t) \sim B(\frac{1}{2}) \\ &\Gamma_\zeta(t+\Delta t) \sim B(\frac{1}{2}) \\ &(\Gamma_d, \Gamma_e)(t+\Delta t) = \begin{cases} (\Gamma_\zeta(t), \Gamma_\zeta(t)) & \text{if } \Gamma_p(t) = 1 \\ (\Gamma_\zeta(t), 1-\Gamma_\zeta(t)) & \text{if } \Gamma_p(t) = 0 \end{cases}, \end{aligned} \tag{16}$$

where the updates of Γ_p and Γ_ζ are independent.

This dynamics provides a stochastic inverse of the XOR operation. Then,

$$\begin{aligned} &T_{p\to d} = T_{p\to e} = 0 \\ &rT_{d\to p} = rT_{e\to p} = 0 \\ &T^2_{p\to de} = -ln\ 2 \\ &rT^2_{de\to p} = -ln\ 2 \\ &T^2_{de\to p} = rT^2_{p\to de} = 0. \end{aligned} \tag{17}$$

Hence, from the sets $V = \{p, \zeta, d, e\}$, $\mathcal{G}_T = (V, \{(\{p\}, \{d, e\})\}, (\{\zeta\}, \{d\}))$, and $\mathcal{G}_{rT} = (V, \{(\{d, e\}, \{p\}), (\{d\}, \{\zeta\})\})$, the information flow hypergraph of this system is defined as $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ (Fig. 1(b)), and satisfies the strong blocking flows condition.

4 Higher-Order Information Dynamics without Weak BF-hypercycles

Under an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ with the strong blocking flows condition, the topology considered below is not required to determine the balance of multiple mutual information. On the other hand, the dynamics of individual entropy H_v, as described in Eq. (8), is determined by higher-order information flows, such as $T^u_{K_v\to v}$ and $rT^u_{K_v\to v}$. From the strong blocking flows condition, a nonzero $rT^u_{K_v\to v}$ implies the nonzero $T^u_{v\to K_v}$. We therefore focus on the hyperarcs associated with nonzero $T^u_{K_v\to v}$ and $T^u_{v\to K_v}$ in $\mathcal{G}_T = (V, \overrightarrow{E}_T)$. These B- and F-arcs form a BF-hypergraph substructure $\mathcal{G}_{BF} = (V, \overrightarrow{E}_{BF})$ of $\mathcal{G}_T$. To clarify the role of cyclic structure, we first examine BF-hypergraphs without weak BF-hypercycles. In classical graph theory [23, 24], a directed graph without cycles has a sink and a source. Regarding a weak BF-hypercycle, a similar argument is suggested [20].

Proposition 3 *A finite BF-hypergraph $\mathcal{G}_{BF}$ without weak BF-hypercycles possesses a BF-sink and a BF-source.*

(proof)

Assume that a BF-sink does not exist in $\mathcal{G}_{BF} = (V, \overrightarrow{E})$. Then, every vertex in V has at least one B-outflow or F-outflow. For a fixed vertex $v_1 \in V$, we consider a weak hyperpath Y_n and a vertex set $V_n \subset V$ such that:

$$\begin{aligned} Y_n &:= (v_1, \overrightarrow{e_1}, v_2, \overrightarrow{e_2}, \ldots, \overrightarrow{e_n}, v_{n+1}) \\ V_n &:= \bigcup_{i=1}^{n} \{v_i\}. \end{aligned} \tag{18}$$

The hyperpath Y_1 exists and $|V_1| = 1$. If Y_n exists, then Y_{n+1} also exists, since v_{n+1} has at least one B-outflow or F-outflow. Meanwhile, if $|V_n| = n$, then $|V_{n+1}| = n+1$ from the fact, $v_{n+1} \notin V_n$, which is required to avoid the formation of a weak BF-hypercycle. However, this contradicts the assumption $|V| < \infty$. The existence of a BF-source follows symmetrically by applying the same argument to the symmetric image of the hypergraph. ■

Therefore, a BF-hypergraph without weak BF-hypercycles in $\mathcal{G}_T$ has a BF-sink and BF-source. The information dynamics of BF-sink is analyzed in Proposition 4 using Lemma 1.

Lemma 1 $T^l_{S\to v} = 0$ *for every nonempty subset* $S \subset K_v$*, where* $l := |S|$*, if and only if* $I_{K_v:v'}|_v = 0$.

(proof)

($\Longrightarrow$) Let $m = |K_v|$. Since,

$$I_{K_v:v'}|_v = \sum_{l=1}^{m} (-1)^{l+1} \sum_{S\subset K_v, |S|=l} T^l_{S\to v}, \tag{19}$$

the result follows.

($\Longleftarrow$) If $I_{K_v:v'}|_v = 0$, $T^1_{S\to v} = 0$ for $\forall S \subset K_v$, such that $|S| = 1$, since,

$$I_{K_v:v'}|_v = T^1_{S\to v} + I_{K_v-S:v'}|_{v,S}, \tag{20}$$

and $T^1_{S\to v} \geqslant 0$ and $I_{K_v-S:v'}|_{v,S} \geqslant 0$.

Now, suppose that, for $n \in \mathbb{N}$, $T^l_{S\to v} = 0$ for $\forall S \subset K_v$, such that $|S| = l \leqslant n$. Regarding an arbitrary set $U \subset K_v$ with $|U| = n+1$, one obtains $I_{U:v'}|_v = 0$, as,

$$I_{K_v:v'}|_v = I_{U:v'}|_v + I_{K_v-U:v'}|_{v,U} = 0, \tag{21}$$

and $I_{U:v'}|_v \geqslant 0$ and $I_{K_v-U:v'}|_{v,U} \geqslant 0$.

Then,

$$I_{U:v'}|_v = \sum_{l=1}^{n} (-1)^{l+1} \sum_{S\subset U, |S|=l} T^l_{S\to v} + (-1)^{n+2} T^{n+1}_{U\to v} = 0. \tag{22}$$

From the assumption, $T^{n+1}_{U\to v} = 0$.

The induction leads to the desired result. ■

Proposition 4 *Consider an information flow hypergraph* $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ *satisfying the strong blocking flows condition, and suppose* $\alpha_1^{K_v\to v} = 0$*. Let* v *be a BF-sink in* $\mathcal{G}_T$*.* ***(a)*** *If* $In_B(v) \neq \emptyset$*, then* $\Delta H_v > 0$*. Otherwise,* ***(b)*** *if* $In_B(v) = \emptyset$*, then* $\Delta H_v = 0$.

(proof)

Consider a BF-sink v with its dynamical neighbors $K_v = \{w_1, \ldots, w_m\}$ in $\mathcal{G}_T$. From Eq. (8) , the dynamics of v's entropy H_v is:

$$\Delta H_v = \sum_{w\in K_v} [T_{w\to v} - rT_{w\to v}] - E_{K_v\to v} + \alpha_1^{K_v\to v}. \tag{23}$$

If a B-inflow $(e^+, \{v\})$ in $\mathcal{G}_{rT}$ is nonzero, the corresponding F-outflow $(\{v\}, e^+)$ in $\mathcal{G}_T$ is also nonzero from the strong blocking flows condition. However, it is not the case, as v is the BF-sink in $\mathcal{G}_T$. Hence, all B-inflows to v in $\mathcal{G}_{rT}$ are zero, resulting in:

$$\begin{aligned} \Delta H_v &= I_{K_v:v'}|_v + \alpha_1^{K_v\to v} \\ &= I_{K_v:v'}|_v \quad (\because \text{Sec.3.3}) \\ &\geqslant 0. \end{aligned} \tag{24}$$

From Lemma 1, if $I_{K_v:v'}|_v = 0$, all B-inflows in $\mathcal{G}_T$ vanish. **(a)** When $In_B(v) \neq \emptyset$, it is not the case, resulting in $\Delta H_v > 0$. **(b)** On the other hand, since $In_B(v) = \emptyset$ indicates $I_{K_v:v'}|_v = 0$ by Lemma 1, one obtains $\Delta H_v = 0$. ■

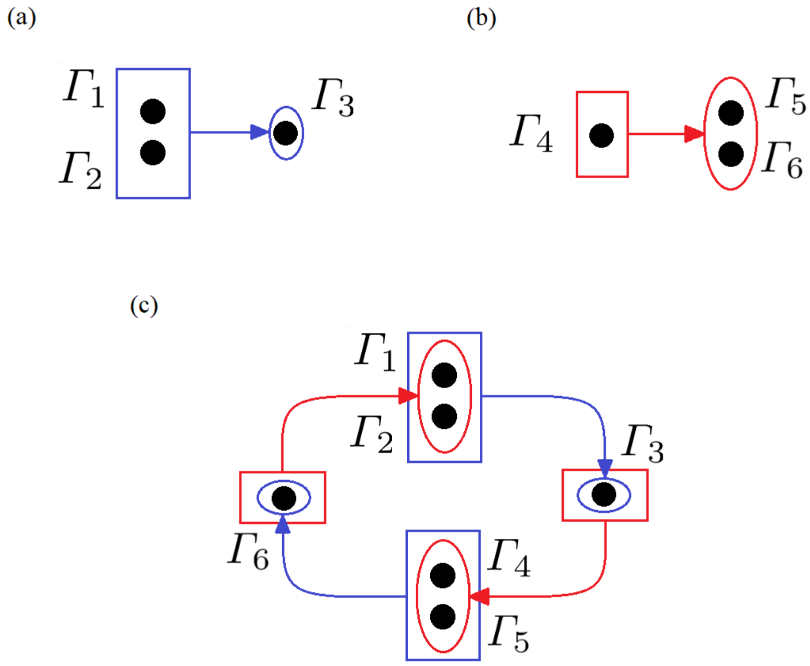

Figure 1: **(a)** The transfer-entropy hypergraph representing the XOR operation: $\Gamma_3' = \Gamma_1 \oplus \Gamma_2$. **(b)** The transfer-entropy hypergraph representing a stochastic inverse of the XOR operation, defined in Eq. (16). **(c)** A weak BF-hypercycle satisfying the necessary condition for stationary information dynamics. The enclosing shapes indicate the source and target sets of each hyperarc. Blue arrows indicate B-arcs, whereas red arrows indicate F-arcs.

Corollary 1 *Consider an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ satisfying the strong blocking flows condition, and suppose $\alpha_1^{K_v \to v} = 0$. Let v be a BF-source in $\mathcal{G}_T$.* ***(a)*** *If $Out_F(v) \neq \emptyset$, then $\Delta H_v < 0$. Otherwise,* ***(b)*** *if $Out_F(v) = \emptyset$, then $\Delta H_v = 0$.*

In conclusion, under the standing assumptions, a nontrivial oscillatory or recurrent state requires the presence of a weak BF-hypercycle.

5 Higher-Order Information Dynamics on a Single Weak BF-hypercycle

To sustain nontrivial oscillatory or recurrent dynamics, an information flow hypergraph should possess a weak BF-hypercycle. A single weak BF-hypercycle has two features distinct from a cycle in classical graph theory. First, it can contain a BF-sink or BF-source. One can consider a simple example with a weak BF-hypercycle $\mathcal{C} = (v_1, (\{v_1\}, \{v_2, s\}), v_2, (\{v_2\}, \{v_3\}), v_3, (\{v_3\}, \{v_1\}), v_1)$, where s is the BF-sink. Nevertheless, a BF-sink or BF-source in a weak BF-hypercycle has constant entropy by Proposition 4 and Corollary 1, since they cannot have B-inflow or F-outflow in the single BF-hypercycle. Therefore, these BF-sinks and BF-sources have no role in the information dynamics of the BF-hypercycle.

Second, a stationary state need not exist for a weak BF-hypercycle. This contrasts with the cycle in classical graph theory, which always has a stationary state with nonzero pairwise transfer entropy and reversed transfer entropy [17]. Lemma 2 shows how the entropy balance is determined by the types of inflow and outflow.

Lemma 2 *Consider an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ satisfying the strong blocking flows condition, and suppose $\alpha_1^{K_v \to v} = 0$. For a vertex v in $\mathcal{G}_{\Delta I}$:*

(a) *$\Delta H_v > 0$ if v has a B-inflow and a B-outflow but no F-outflow in $\mathcal{G}_T$.*

(b) *$\Delta H_v < 0$ if v has an F-inflow and an F-outflow but no B-inflow in $\mathcal{G}_T$.*

(c) *$\Delta H_v = 0$ if v has an F-inflow and a B-outflow but neither a B-inflow nor an F-outflow in $\mathcal{G}_T$.*

(d) *ΔH_v may be positive, negative, or zero if v has both a B-inflow and an F-outflow in $\mathcal{G}_T$.*

(proof)

(a) Let v have only B-inflow $(P, \{v\})$ and B-outflow $(\{u_1, \ldots, u_l, v\}, \{s\})$ in $\mathcal{G}_T$. The strong blocking flows condition requires that F-outflow $(\{v\}, P)$ and F-inflow $(\{s\}, \{u_1, \ldots, u_l, v\})$ exist in $\mathcal{G}_{rT}$. Then, from Lemma 1, $I_{K_v:v'}|_v \neq 0$ and $I_{K_v':v}|_{v'} = 0$, where $K_v' := \{w'|w \in K_v\}$. Hence,

$$\begin{aligned} \Delta H_v &= I_{K_v:v'}|_v - I_{K_v':v}|_{v'} + \alpha_1^{K_v \to v} \\ &= I_{K_v:v'}|_v \quad (\because \text{Sec.3.3}) \\ &> 0 \quad (\because I_{K_v:v'}|_v \neq 0). \end{aligned} \tag{25}$$

(b) Let v have only F-inflow $(\{s\}, \{u_1, \ldots, u_l, v\})$ and F-outflow $(\{v\}, Q)$ in $\mathcal{G}_T$. The strong blocking flows condition requires that B-outflow $(\{u_1, \ldots, u_l, v\}, \{s\})$ and B-inflow $(Q, \{v\})$ exist in $\mathcal{G}_{rT}$. Then, from Lemma 1, $I_{K_v:v'}|_v = 0$ and $I_{K_v':v}|_{v'} \neq 0$, where $K_v' := \{w'|w \in K_v\}$. Hence,

$$\begin{aligned} \Delta H_v &= I_{K_v:v'}|_v - I_{K_v':v}|_{v'} + \alpha_1^{K_v \to v} \\ &= -I_{K_v':v}|_{v'} \quad (\because \text{Sec.3.3}) \\ &< 0 \quad (\because I_{K_v':v}|_{v'} \neq 0). \end{aligned} \tag{26}$$

(c) Let v have only F-inflow $(\{s\}, \{u_1, \ldots, u_l, v\})$ and B-outflow $(\{w_1, \ldots, w_k, v\}, \{p\})$ in $\mathcal{G}_T$. The strong blocking flows condition requires that B-outflow $(\{u_1, \ldots, u_l, v\}, \{s\})$ and F-inflow $(\{p\}, \{w_1, \ldots, w_k, v\})$ exist in $\mathcal{G}_{rT}$. Then, from Lemma 1, $I_{K_v:v'}|_v = 0$ and $I_{K_v':v}|_{v'} = 0$, where $K_v' := \{w'|w \in K_v\}$. Hence,

$$\begin{aligned} \Delta H_v &= I_{K_v:v'}|_v - I_{K_v':v}|_{v'} + \alpha_1^{K_v \to v} \\ &= 0 \quad (\because \text{Sec.3.3}). \end{aligned} \tag{27}$$

(d) Let v have only B-inflow $(P, \{v\})$ and F-outflow $(\{v\}, Q)$ in $\mathcal{G}_T$. The strong blocking flows condition requires that F-outflow $(\{v\}, P)$ and B-inflow $(Q, \{v\})$ exist in $\mathcal{G}_{rT}$. Then, from Lemma 1, $I_{K_v:v'}|_v \neq 0$ and $I_{K_v':v}|_{v'} \neq 0$, where K_v is a set of dynamical neighbors, and $K_v' := \{w'|w \in K_v\}$. Hence,

$$\begin{aligned} \Delta H_v &= I_{K_v:v'}|_v - I_{K_v':v}|_{v'} + \alpha_1^{K_v \to v} \\ &= I_{K_v:v'}|_v - I_{K_v':v}|_{v'} \quad (\because \text{Sec.3.3}). \blacksquare \end{aligned} \tag{28}$$

Proposition 5 *Consider an information flow hypergraph $\mathcal{G}_{\Delta I} = (\mathcal{G}_T, \mathcal{G}_{rT})$ satisfying the strong blocking flows condition, and suppose $\alpha_1^{K_v \to v} = 0$. A necessary condition for stationarity is that, for every vertex v in $\mathcal{G}_{\Delta I}$, v has a B-inflow in $\mathcal{G}_T$ if and only if it has an F-outflow in $\mathcal{G}_T$.*

(proof)

Stationarity requires $\Delta H_v = 0$ for every vertex v in $\mathcal{G}_{\Delta I}$. Therefore, the necessary condition for the stationary state is obtained by excluding the monotonic cases. From Proposition 4 and Corollary 1, v cannot be a BF-sink with B-inflow and a BF-source with F-outflow. Furthermore, Lemma 2 shows that ΔH_v cannot be zero, when v has a B-inflow but no F-outflow in $\mathcal{G}_T$, or has an F-outflow but no B-inflow in $\mathcal{G}_T$.

In summary, if v has a B-inflow in $\mathcal{G}_T$, v should have an outflow not to be a BF-sink with B-inflow, and the outflow should contain an F-outflow in $\mathcal{G}_T$. Similarly, if v has an F-outflow in $\mathcal{G}_T$, v should have an inflow not to be a BF-source with F-outflow, and the inflow should contain a B-inflow in $\mathcal{G}_T$. ■

A cycle in classical graph theory satisfies the necessary condition of Proposition 5, as the directed edge of the cycle is both B-arc and F-arc.

Furthermore, one can construct a simple example for Proposition 5 by using the B-arc of Sec. 3.4 and the F-arc of Sec. 3.5. Figure 1(c) represents a circuit consisting of an XOR operation and a stochastic inverse of the XOR operation. At every vertex in the circuit, $\alpha_1 = 0$ holds, though the dynamics is not strictly deterministic and reversible. The information dynamics of the circuit can be stationary, satisfying Proposition 5.

6 Conclusion

Higher-order interactions can generate higher-order correlations that are not reducible to pairwise relationships. To characterize the topology of their temporal evolution, this paper introduced an information flow hypergraph by associating direct multiple transfer entropy and multiple reversed transfer entropy with hyperarcs. The BF-hypergraph substructure determines the entropy balance of individual vertices, under the strong blocking flows condition and $\alpha_1^{K_v \to v} = 0$. A finite BF-hypergraph without weak BF-hypercycles possesses a BF-sink and a BF-source. At a BF-sink, the entropy is nondecreasing and increases strictly in the presence of a B-inflow. Consequently, nontrivial oscillatory or recurrent information dynamics requires a weak BF-hypercycle under these assumptions. However, a weak BF-hypercycle alone does not guarantee stationarity. A further necessary condition is that, at each vertex, a B-inflow is present if and only if an F-outflow is present. This condition also clarifies the role of irreversibility. In a stationary state satisfying the strong blocking flows condition, an isolated B-inflow without a compensating F-outflow requires $\alpha_1 \neq 0$ and is therefore incompatible with component-wise reversible dynamics. The XOR operation provides a simple example of such an irreversible higher-order transformation.

The present analysis focuses on sinks, sources, and weak BF-hypercycles. Directed hypergraphs possess additional structural properties, including spectral properties of hypergraph Laplacians [34], that may provide further insight into information dynamics. Another natural extension is to study the relationship between an explosive behavior of higher-order dynamics [1] and information-dynamic stationarity discussed in this paper.

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A Proof of Proposition 1

A.1 Main Proof

One can establish a non-trivial example involving a common cause. Define $V = \{X, Y, C, \zeta\}$ and their states Γ_X, Γ_Y, Γ_C, Γ_ζ, where $\Gamma_C = (\Gamma_{C_1}, \Gamma_{C_2})$. Their dynamics is defined as:

$$\begin{aligned}
&\Gamma_X(t+\Delta t) = \Gamma_{C_1}(t) \\
&\Gamma_Y(t+\Delta t) = \Gamma_{C_2}(t) \\
&\Gamma_{C_1}(t+\Delta t) = \Gamma_{C_2}(t) \\
&\Gamma_{C_2}(t+\Delta t) = \Gamma_\zeta(t) \\
&\Gamma_\zeta(t+\Delta t) \sim B(\frac{1}{2}),
\end{aligned} \tag{29}$$

where $B(q)$ denotes Bernoulli distribution with probability q.

Then, $K_X = K_Y = \{C\}$. While $T_{Y\to X} = ln\ 2$, X is not connected to Y in $\mathcal{G}_T$. Hence, $\mathcal{G}_T^* \neq \mathcal{G}_{total}^*$ in this example. ■

The subsequent sections explain why the common-cause case is a natural source of indirect statistical relationships that are not represented by chains in $\mathcal{G}_T$.

A.2 Total Neighbors

For a vertex set $S \subset V$, let $Q_S^1 \subset V$ be the union of possible dynamical neighbors of S: $Q_S^1 := \bigcup_{K_S \in \overline{\mathcal{K}}_S} K_S$. Then, define $Q_S^{i+1} \subset V$ as the union of possible dynamical neighbors of $Q_S^i \subset V$ for $i \in \mathbb{N}$: $Q_S^{i+1} := \bigcup_{K_{Q_S^i} \in \overline{\mathcal{K}}_{Q_S^i}} K_{Q_S^i}$. Since V is finite, there is $N \in \mathbb{N}$ such that,

$$\bigcup_{i=1}^{N} Q_S^i = \bigcup_{i=1}^{N+1} Q_S^i. \tag{30}$$

Hence, the set of total neighbors of S is defined as:

$$K_S^{total} := \bigcup_{i=1}^{N} Q_S^i. \tag{31}$$

Lemma 3 *If $u \in K_S^{total}$, then $\exists s \in S$ such that s is connected to u in $\mathcal{G}_T$.*

(proof)

It is enough to show that, for $u \in K_S$ for $K_S \in \overline{\mathcal{K}}_S$, $\exists (e^+, e^-) \in \overrightarrow{E}_T$ such that $u \in e^+$ and $e^- \cap S \neq \emptyset$. Let $K_S = \{u, w_1, \dots, w_l\}$, and $S = \{s_1, \dots, s_m\}$, and $(S) = (s_1, \dots, s_m)$. By the definition of K_S,

$$H_{(S')}|_{(S),w_1,\dots,w_l} > 0. \tag{32}$$

Then, there is $s \in S$ such that $H_{s'}|_{(S),w_1,\dots,w_l} > 0$, as,

$$\begin{aligned} H_{(S')}|_{(S),w_1,\dots,w_l} &= H_{s'_1}|_{(S),w_1,\dots,w_l} + \sum_{i=2}^{m} H_{s'_i}|_{s'_1,\dots,s'_{i-1},(S),w_1,\dots,w_l} \\ &\leqslant \sum_{i=1}^{m} H_{s'_i}|_{(S),w_1,\dots,w_l}. \end{aligned} \tag{33}$$

Let $L_s := [S - \{s\}] \cup [K_S - \{u\}]$. As $H_{s'}|_{s,u,(L_s)} \leqslant H_{(S')}|_{s,u,(L_s)} = 0$, $H_{s'}|_{s,u,(L_s)} = 0$. Thus, one can obtain a minimal subset $K_s \subset L_s \cup \{u\}$ such that $H_{s'}|_{s,(K_s)} = 0$, by reducing the set $L_s \cup \{u\}$. Since $H_{s'}|_{s,(L_s)} > 0$, $u \in K_s$. Then,

$$\begin{aligned} T_{u \to s}|_{(K_s - \{u\})} &:= H_{s'}|_{s,(K_s - \{u\})} - H_{s'}|_{s,(K_s)} \\ &= H_{s'}|_{s,(K_s - \{u\})} > 0. \end{aligned} \tag{34}$$

Since,

$$\begin{aligned} T_{u \to s}|_{(K_s - \{u\})} &:= T_{u \to s} + I^2_{u:(K_s - \{u\}):s'}|_s \\ &= T_{u \to s} + \sum_{i=1}^{|K_s|-1} (-1)^{i+1} \sum_{W \subset K_s - \{u\}, |W| = i} T^{i+1}_{\{u\} \cup W \to s}, \end{aligned} \tag{35}$$

there is a nonzero term $T^k_{e^+ \to s}$ among $T_{u \to s}$, and $T^{i+1}_{\{u\} \cup W \to s}$ with $k := |e^+|$. One can confirm $e^+ \subset K_s$. Hence, the hyperarc $(e^+, \{s\})$ is an element of $\overrightarrow{E}_T$. Given that $u \in e^+$, s is connected to u in $\mathcal{G}_T$.

If $u \in K_S^j$ for $K_S^j \in Q_S^j$ and $j \in \mathbb{N}$, $j \leqslant N$, the inductive argument proves the claims. ■

The state dynamics of $S^{total} := S \cup K_S^{total}$ is autonomous: $\Gamma_{S^{total}}(t + \Delta t) = F(\Gamma_{S^{total}}(t), t)$, where $\Gamma_{S^{total}}(t) := (\Gamma_{w_1}(t), \dots, \Gamma_{w_h}(t))$ with $S^{total} = \{w_1, \dots, w_h\}$.

A.3 Two Independent Autonomous Systems

Consider two total neighbors S^{total} and U^{total} such that $S^{total} \cap U^{total} = \emptyset$.

Lemma 4 *If $I^{m+k-1}_{s_1:\cdots:s_m:u_1:\cdots:u_k}(t_0) = 0$ for $\forall s_1, \dots, s_m \in S^{total}$ and $\forall u_1, \dots, u_k \in U^{total}$, then $I^{m+k-1}_{s_1:\cdots:s_m:u_1:\cdots:u_k}(t) = 0$ and $T^{m+k-1}_{\{s_1,\dots,s_m\} \to \{u_1,\dots,u_k\}}(t) = 0$ for $t \geqslant t_0$.*

(proof)

Since S^{total} and U^{total} are autonomous, one obtains,

$$\begin{aligned} \Gamma_{S^{total}}(t) &= F_t(\Gamma_{S^{total}}(t_0)) \\ \Gamma_{U^{total}}(t') &= G_{t'}(\Gamma_{U^{total}}(t_0)), \end{aligned} \tag{36}$$

for $t, t' \geqslant t_0$.

On the other hand, if $I^{m+k-1}_{s_1:\cdots:s_m:u_1:\cdots:u_k}(t_0) = 0$ for $\forall s_1, \dots, s_m \in S^{total}$ and $\forall u_1, \dots, u_k \in U^{total}$, then $I_{(S^{total}):(U^{total})}(t_0) = 0$. This indicates the factorization of joint probability.

As $S^{total} \cap U^{total} = \emptyset$ and statistically independent at $t = t_0$, $\Gamma_{S^{total}}(t)$ and $\Gamma_{U^{total}}(t')$ are statistically independent for $\forall t, t' \geqslant t_0$. ■

A.4 Classification of Indirect Relationships

We analyze case by case whether S is connected to U in $\mathcal{G}_T$ for $\forall (U, S) \in \overrightarrow{E}_{total}$. Fix an arbitrary hyperarc $(U, S) \in \overrightarrow{E}_{total}$. If $U \cap K_S^{total} \neq \emptyset$, Lemma 3 requires that S is connected to U in $\mathcal{G}_T$. Let $U \cap K_S^{total} = \emptyset$.

If $U^{total} \cap S^{total} = \emptyset$, Lemma 4 implies that (U, S) cannot be an element of $\overrightarrow{E}_{total}$ under the condition of initial independence.

Let $U^{total} \cap S^{total} \neq \emptyset$. Here, one can identify the element of $U^{total} \cap S^{total}$ as a common cause addressed in Sec. A.1.

B The Proof of Proposition 2

Since $\overrightarrow{E}_{\bar{T}} \subset \overrightarrow{E}_{total}$, it follows immediately that $\mathcal{G}^*_{\bar{T}} \subset \mathcal{G}^*_{total}$. It therefore remains to show that $\mathcal{G}^*_{total} \subset \mathcal{G}^*_{\bar{T}}$.

Consider $(U, S) \in \overrightarrow{E}_{total}$, and fix $K_S \in \overline{\mathcal{K}}_S$. Then, $T^{l+m-1}_{U\to S} \neq 0$, where $l = |U|$ and $m = |S|$. If $U \cap K_S \neq \emptyset$, then $(U, S) \in \overrightarrow{E}_{\bar{T}}$ by definition. Let $U \cap K_S = \emptyset$. The decomposition of $T^{l+m-1}_{U\to S}|_{K_S}$ is obtained as:

$$T^{l+m-1}_{U\to S}|_{K_S} = T^{l+m-1}_{U\to S} + \sum_{i=1}^{|K_S|} (-1)^{i+1} \sum_{W\subset K_S, |W|=i} T^{l+m+i-1}_{U\cup W\to S}. \tag{37}$$

Since $T^{l+m-1}_{U\to S}|_{K_S} = 0$ and $T^{l+m-1}_{U\to S} \neq 0$, there is a nonzero term $T^{l+m+i-1}_{U\cup W\to S}$ for $\emptyset \neq W \subset K_S$ and $i = |W|$. Therefore, the hyperarc $(U \cup W, S) \in \overrightarrow{E}_{total}$, and one concludes that $(U \cup W, S) \in \overrightarrow{E}_{\bar{T}}$ from the fact that $(U \cup W) \cap K_S = W \neq \emptyset$. This leads to $\mathcal{G}^*_{total} \subset \mathcal{G}^*_{\bar{T}}$. ■